

Evaluation report 1: [TheMarker](#)

Arthur Issler

Source URL(s): <https://themarker.idrblab.cn/>

Maintainer/Organization: TheMarker is maintained by IDRBLab at the College of Pharmaceutical Sciences, Zhejiang University.

Last updated (if available): Not found. The associated publication was released in 2024, but I could not confirm when the database itself was last updated.

Description / what it covers:

TheMarker is a database of therapeutic biomarkers. It focuses on biomarkers related to drug treatment and clinical response rather than mainly diagnostic or prognostic biomarkers.

It includes five therapeutic biomarker categories:

- Pharmacodynamic biomarkers
- Safety biomarkers
- Monitoring biomarkers
- Predictive biomarkers
- Surrogate endpoints

Records may connect a biomarker with a disease, treatment, specimen, testing method, and supporting results.

Access method (API, bulk download, scrape, manual):

The data can be accessed through:

- Search and browse pages on the website
- Individual biomarker, drug, and disease pages
- Bulk downloads for the full database and individual biomarker categories

I did not find documentation for a stable public API. The website uses URL parameters for searches, but these should not be treated as an official API without confirmation.

License:

The associated publication is licensed under Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial

re-use, distribution, and reproduction in any medium, provided the original work is properly cited. But, I could not find a separate license that clearly applies to the downloadable database. The data reuse terms should therefore be confirmed with the maintainers before integration.

Field mapping: can it populate the critical fields and the rest of the schema?:

Field	Support	Notes
Source record ID/persistent URL (biomarker_id)	Yes	Records have internal identifiers, such as BS3PC9.
biomarker (CRITICAL)	Partial	Biomarker names are provided, but the complete biomarker statement may need to be built from the name, result, and direction of change.
assessed_biomarker_entity (CRITICAL)	Yes	Gene, protein, RNA, and other biomarker entity names are provided.
assessed_biomarker_entity_id (CRITICAL)	Partial	Most records include external identifiers such as NCBI Gene, UniProtKB, or HGNC IDs, but at least one reviewed record had no external links. Overall coverage has not been measured.
assessed_entity_type	Yes	The biomarker class may identify types such as protein, mRNA, or microRNA.
condition (CRITICAL)	Yes	Diseases or treatment indications are included in the records.
condition_id	Mappable	Some conditions use ICD identifiers. These would need to be mapped to DOID or MONDO for BiomarkerKB.
exposure_agent	Yes	Drugs and therapies are central parts of the database.
exposure_agent_id	Partial	Internal drug IDs are available, but I could not confirm consistent external drug identifiers.
best_biomarker_role	Partial	The source provides therapeutic roles such as predictive, safety, and monitoring. These do not perfectly match BiomarkerKB's current role categories.

specimen	Partial	Tissue or sample information is available for some records and transcriptomic datasets, but consistent record-level specimen coverage has not been measured.
specimen_id	Mappable	Specimens appear to be mainly written as text and would need mapping to UBERON or another ontology.
loinc_code	No	I found no evidence that LOINC codes are provided.
evidence_source	Partial	Some records include publications or datasets such as GEO accessions, but evidence sources may not be consistent across all records.
evidence	Partial	Records may include result descriptions, fold changes, p-values, testing methods, or other supporting information. The format varies.

Data quality / standardization:

TheMarker appears to combine literature review, manual curation, and analysis of transcriptomic datasets. It uses structured disease codes and includes external identifiers for many biomarker entities.

However, the data do not directly match the BiomarkerKB schema. Biomarker statements may need to be assembled from several fields, and specimen and evidence information may require cleaning and normalization.

Data volume/scope estimate:

The associated publication reports approximately 16,942 therapeutic biomarker records:

- 218 pharmacodynamic biomarkers
- 624 safety biomarkers
- 104 monitoring biomarkers
- 15,893 predictive biomarkers
- 103 surrogate endpoints

The database covers hundreds of drugs and many disease categories. These counts should be treated as estimates until the current download files are checked. ([Source](#))

Recommended integration status: Sample integration

TheMarker appears able to provide the critical biomarker, assessed entity, and condition fields. It could also add valuable treatment-response, drug, toxicity, monitoring, and surrogate endpoint information.

A sample integration is more appropriate than direct integration because:

- The database license is unclear.
- Biomarker statements may need to be reconstructed.
- Its therapeutic roles do not perfectly match BiomarkerKB's roles.
- Disease and specimen identifiers require mapping.
- A stable public API was not found.

A pilot using a small number of predictive or monitoring biomarker records could show how well the data fit BiomarkerKB.

Open questions / limitations / blockers:

- Does the database license allow its downloadable data to be reused in BiomarkerKB?
- Are all fields included in the bulk downloads, or only general biomarker information?
- How consistently are external biomarker and drug identifiers provided?
- How should the five therapeutic roles map to BiomarkerKB's role vocabulary?
- How difficult would it be to map ICD-11 conditions to DOID or MONDO?
- How consistently are specimens and evidence sources recorded?

Field	Test/Example Filing: 1: TheMarker https://themarker.idrblab.cn/data/marker?id=B1OZJ0
Source record ID / persistent URL (biomarker id)	B1OZJ0
biomarker (CRITICAL)	increased expression of RNA SERPINB4/HGNC:10570
assessed_biomarker_entity (CRITICAL)	SERPINB4
assessed_biomarker_entity_id (CRITICAL)	HGNC:10570
assessed entity type	mRNA
condition (CRITICAL)	Pancreatic cancer
condition_id	

exposure agent	fluorouracil
exposure agent id	
best biomarker role	predictive
specimen	sample from patient
specimen id	
loinc code	
evidence source	TCGA PAAD
evidence	Non response vs Response; log2FC: 8.81E+00; P-adj: 1.52E-02; P-value: 7.03E-04

Evaluation Report 2: ResMarkerDB

Arthur Issler

Source URL(s): Original database: <http://www.resmarkerdb.org>

Publication: Judith Pérez-Granado, Janet Piñero, Laura I Furlong, ResMarkerDB: a database of biomarkers of response to antibody therapy in breast and colorectal cancer, Database, Volume 2019, 2019, baz060, <https://doi.org/10.1093/database/baz060>

Maintainer/Organization: ResMarkerDB was developed by researchers at Pompeu Fabra University and the Hospital del Mar Medical Research Institute in Barcelona, Spain.

Last updated (if available): Not found. The database was described in a 2019 publication, but the website is no longer accessible.

Description / what it covers:

ResMarkerDB collected biomarkers linked to treatment sensitivity or resistance in breast and colorectal cancer.

It included biomarkers such as:

- Gene variants
- Copy number changes
- Gene and protein expression
- microRNAs and long non-coding RNAs

Records connected biomarkers with cancer types, treatments, treatment response, and supporting evidence.

Access method (API, bulk download, scrape, manual):

The database originally had a searchable website and downloadable data. The website is currently unavailable, and I could not find a replacement website, public download, API, or code repository. The publication and supplementary files are the only accessible sources I found.

License:

The publication states that ResMarkerDB was available under a Creative Commons 4.0 license. However, some records came from databases with separate restrictions, particularly JAX-CKB. The licensing of any recovered dataset would need to be reviewed before reuse.

Field mapping: can it populate the critical fields and the rest of the schema?:

Field	Support	Notes
Source record ID/persistent URL (biomarker_id)	Unknown	Internal record identifiers could not be confirmed without access to the database
biomarker (CRITICAL)	Yes	The paper describes specific alterations such as mutations, amplification, and overexpression.
assessed_biomarker_entity (CRITICAL)	Yes	Gene, protein, miRNA, and lncRNA entities were included.
assessed_biomarker_entity_id (CRITICAL)	Partial	The paper reports use of identifiers such as NCBI Gene and dbSNP, but full coverage cannot be checked.
assessed_entity_type	Yes	Several biomarker types were described.
condition (CRITICAL)	Yes	Breast cancer, colorectal cancer, and cancer subtypes were included.
condition_id	Yes	Cancer terms were mapped to resources including NCIT and Disease Ontology.
exposure_agent	Yes	Drugs and treatment combinations were central to the records.
exposure_agent_id	Partial	DrugBank and ATC identifiers were used, but coverage cannot be checked.
best_biomarker_role	Yes	The biomarkers were predictive of treatment sensitivity or resistance.
specimen	Unknown	Specimen fields could not be confirmed.
specimen_id	Unknown	No specimen ontology identifiers were described.
loinc_code	Unknown	No LOINC support was described.
evidence_source	Yes	Records were connected to publications, databases, guidelines, or clinical evidence.
evidence	Yes	The paper states that supporting evidence statements were included.

Data quality / standardization:

The paper describes the use of controlled vocabularies, external identifiers, and expert review. This suggests that the database was well structured. However, the records cannot currently be inspected, so data completeness and consistency cannot be independently checked.

Data volume/scope estimate:

The 2019 publication reported:

- 266 biomarkers
- More than 500 biomarker–drug–cancer associations
- 134 treatments
- 73 individual drugs

These numbers only describe the version published in 2019.

Recommended integration status: Not viable

Not viable at present - ResMarkerDB appears relevant to BiomarkerKB and likely supports the critical fields. However, the website and data downloads are unavailable, so the records cannot currently be reviewed or integrated. If the maintainers provide the original dataset, the source could be reconsidered for sample integration.

Open questions / limitations / blockers:

- Is ResMarkerDB permanently discontinued?
- Can the maintainers provide a database export?
- Is an archived download available?
- What license applies to the complete dataset?

Field	Test/Example Filing 2: ResMarkerDB https://academic.oup.com/database/article/doi/10.1093/database/baz060/5510694
Source record ID / persistent URL (biomarker id)	
biomarker (CRITICAL)	presence of EGFR R451C mutation
assessed_biomarker_entity (CRITICAL)	EGFR
assessed_biomarker_entity_id (CRITICAL)	
assessed entity type	gene mutation
condition (CRITICAL)	colorectal cancer
condition id	
exposure_agent	cetuximab
exposure_agent id	
best biomarker role	predictive
specimen	
specimen id	
loinc code	
evidence source	DOI: 10.1093/database/baz060
evidence	Sensitivity to treatment at preclinical level; resistance to treatment at clinical level

Exact entry used:

<https://academic.oup.com/database/article/doi/10.1093/database/baz060/5510694> — see the Results section around Figure 4, where the paper specifically discusses the EGFR R451C–cetuximab–colorectal cancer combination.

Notes: The original ResMarkerDB database is not currently accessible, so this example comes directly from the original paper rather than an individual database entry. I used predictive because the example specifically describes whether the mutation is associated with sensitivity or resistance to cetuximab treatment.

Evaluation report 3: SalivaDB

Arthur Issler

Source URL(s):

Database: <https://webs.iiitd.edu.in/raghava/salivadb/>

Publication: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9902669/>

GitHub: <https://github.com/raghavagps/SalivaDB>

Zenodo: <https://doi.org/10.5281/zenodo.20078761>

Maintainer/Organization: Developed by the Raghava Group in the Department of Computational Biology at Indraprastha Institute of Information Technology Delhi. Prof. G.P.S. Raghava is listed as the project supervisor and contact.

Last updated (if available): The database includes literature collected through February 3, 2022. An official GitHub and Zenodo package was released on May 8, 2026, but I found no evidence that the biomarker data themselves were updated after 2022.

Description / what it covers:

SalivaDB is a manually curated database of human salivary biomarkers. It includes:

Proteins; Metabolites; Microbes; miRNAs and Genes.

Records connect salivary biomarkers to diseases or conditions and may include diagnostic or prognostic roles, regulation, fold change, specimen collection details, analysis methods, external identifiers, and supporting publications.

Access method (API, bulk download, scrape, manual):

The database can be accessed through:

Basic and advanced search; Browse pages; Individual record pages; TSV downloads for each biomarker category; A downloadable GitHub and Zenodo repository package

I did not find documentation for a public API. Bulk downloads would be preferable to scraping.

License:

The GitHub repository includes an MIT license, while the publication is CC BY 4.0. The license applying to the complete database export, particularly records imported from HMDB and SalivaTecDB, should still be confirmed.

Field mapping: can it populate the critical fields and the rest of the schema?:

Field	Support	Notes
Source record ID/persistent URL (biomarker_id)	Yes	Each record has an internal SAL_ID, such as SAL_11467.
biomarker (CRITICAL)	Partial	Biomarker names are provided, but the complete statement may need to combine the name, regulation, fold change, specimen, and condition. Some records lack regulation information.
assessed_biomarker_entity (CRITICAL)	Yes	Protein, gene, miRNA, metabolite, and microbe names are provided.
assessed_biomarker_entity_id (CRITICAL)	Partial	Records may link to UniProtKB, PubChem, miRBase, NCBI Gene, or NCBI Taxonomy, but some records have no external identifier.
assessed_entity_type	Yes	Each record is classified as a protein, metabolite, microbe, miRNA, or gene.
condition (CRITICAL)	Yes	Disease or condition names are included.
condition_id	Mappable	Disease names are provided, but I found no DOID or MONDO identifiers.
exposure_agent	No	I found no drug or exposure-agent field.
exposure_agent_id	No	No exposure-agent identifiers were found.
best_biomarker_role	Partial	Records may be diagnostic, prognostic, or both, but some records have no assigned role.
specimen	Yes	Saliva is the main specimen, with more specific values such as whole saliva or parotid saliva.

specimen_id	Mappable	Specimen information is mainly text and would need mapping to UBERON or another ontology.
loinc_code	No	I found no evidence of LOINC codes.
evidence_source	Partial	Many records include PMIDs, titles, and abstracts, but some imported records do not contain a standard PMID.
evidence	Partial	Records may contain methods, fold changes, regulation, concentration, and abstracts, but not always a specific supporting text excerpt.

Data quality / standardization:

SalivaDB contains structured fields and was partly manually curated. The publication reports that 6,894 records were manually collected from 478 PubMed articles.

However, the data also include records imported from SalivaTecDB and HMDB. Field completeness varies, some records contain NA values, and disease and specimen terms are not mapped to BiomarkerKB's preferred ontologies.

Data volume/scope estimate:

SalivaDB reports:

- 15,821 total entries
- 7,729 unique biomarkers
- 201 diseases
- 48 disease categories
- 6,067 protein entries
- 3,987 metabolite entries
- 2,909 microbe entries
- 2,272 miRNA entries
- 586 gene entries

Of the total records, 6,894 were manually curated, 5,776 came from SalivaTecDB, and 3,151 came from HMDB.

Recommended integration status: Sample integration

SalivaDB supports the three critical fields and provides downloadable data. It would also expand BiomarkerKB's coverage of saliva as a non-invasive specimen and add several biomarker entity types.

A sample integration is recommended because:

- Some records are incomplete.
- Biomarker statements may need to be reconstructed.
- Disease and specimen terms require ontology mapping.
- Some records describe healthy reference values rather than disease biomarkers.
- Licensing and provenance should be reviewed for records imported from HMDB and SalivaTecDB.

A pilot could begin with manually curated, disease-associated records that include an external entity ID, biomarker role, PMID, and regulation or concentration information.

Open questions / limitations / blockers:

- Does the MIT License apply to all deposited data or mainly the repository files and code?
- What reuse terms apply to records imported from HMDB and SalivaTecDB?
- Were any biomarker records updated when the repository was released in May 2026?
- How many records lack external biomarker identifiers?
- How many records lack a diagnostic or prognostic role?
- How should healthy reference records be handled?
- How consistently can disease and specimen terms be mapped to DOID, MONDO, and UBERON?
- Do the TSV files contain all information shown on the individual record pages?

Field	Test/Example Filing 3: SalivaDB https://webs.iiitd.edu.in/raghava/salivadb/display_saliva2.php?details=SAL_11324
Source record ID / persistent URL (biomarker id)	SAL_11324
biomarker (CRITICAL)	increased level of protein Interleukin-6/UPKB:P05231
assessed_biomarker_entity (CRITICAL)	Interleukin-6
assessed_biomarker_entity_id (CRITICAL)	UPKB:P05231
assessed entity type	protein
condition (CRITICAL)	oral cancer
condition id	
exposure_agent	
exposure_agent id	
best_biomarker role	diagnostic
specimen	whole saliva
specimen id	
loinc code	
evidence source	PMID:31350970
evidence	Upregulated; measured by ELISA

Notes: I changed SalivaDB's “**Upregulated**” into the BiomarkerKB-style wording “**increased level of protein...**”. I left the condition ID, specimen ID, LOINC, and exposure fields blank because this SalivaDB entry does not provide them.

Evaluation report 4: GlycanAge

Arthur Issler

Source URL(s):

- <https://glycanage.com/resources/publications>
- <https://glycanage.com/for-research/glycanage-for-research>
- <https://glycanage.com/legal/terms-and-conditions>

Maintainer/Organization: The publications page is maintained by GlycanAge Ltd. The research behind GlycanAge is connected to the Genos Glycoscience Research Laboratory.

Last updated (if available): Not stated. The page includes publications from 2026, so it appears to be actively maintained.

Description / what it covers:

GlycanAge Publications is a curated list of research papers rather than a biomarker database.

The publications mainly cover:

- IgG and plasma glycan patterns
- Biological and chronological aging
- Inflammation
- Disease risk and prognosis
- Treatment and lifestyle effects on glycosylation

The papers include research on cancer, cardiovascular disease, diabetes, kidney disease, autoimmune conditions, obesity, menopause, and aging.

Access method (API, bulk download, scrape, manual):

The source can be accessed manually through the publications webpage and links to individual papers. I did not find: A biomarker database; A bulk data download; A CSV or TSV export; A public biomarker API; or Structured biomarker records. Biomarkers would need to be manually extracted from selected publications.

License:

I did not find an open data license for the publications page. GlycanAge's website content is protected by its terms and conditions. Each linked publication has its own license. Some articles

are open access, while others may have publisher restrictions. Licensing would therefore need to be checked for each paper used.

Field mapping: can it populate the critical fields and the rest of the schema?:

Field	Support	Notes
Source record ID/persistent URL (biomarker_id)	No	The source does not provide structured biomarker records or record IDs.
biomarker (CRITICAL)	Derived	Biomarker statements can be manually extracted from individual papers, such as increased agalactosylated IgG glycans.
assessed_biomarker_entity (CRITICAL)	Derived	The assessed entities are usually IgG glycans, plasma glycans, glycan structures, or derived glycan traits.
assessed_biomarker_entity_id (CRITICAL)	Mappable	The publications page does not provide glycan IDs. Exact structures may sometimes be mapped to resources such as GlyTouCan, but many reported traits represent groups of glycans.
assessed_entity_type	Derived	Most relevant records would use glycan, although some findings are glycan indices or grouped traits.
condition (CRITICAL)	Partial	Many papers focus on diseases or outcomes, while others study age or healthy populations.
condition_id	Mappable	Conditions would need to be mapped manually to DOID or MONDO.
exposure_agent	Partial	Some studies examine medications, hormones, diets, weight loss, or other interventions.
exposure_agent_id	Mappable	Agents would need to be identified and mapped from each paper.
best_biomarker_role	Partial	Papers may support prognostic, predictive, monitoring, or aging-related roles, but these are not assigned in a standardized format.

specimen	Partial	Papers may use plasma, serum, blood, or dried blood spots, but specimen information is not included on the publications page as a structured field.
specimen_id	Mappable	Specimens would need to be mapped to UBERON or another ontology.
loinc_code	No	I found no evidence of LOINC codes.
evidence_source	Yes	The source links directly to publications, which can provide PMIDs, DOIs, and other citations.
evidence	Partial	Supporting evidence is available within the papers but must be manually extracted from the results, tables, or abstracts.

Data quality / standardization:

The source links to a large amount of relevant scientific research, including peer-reviewed publications and other research outputs, including preprints, conference abstracts, and patents. The scientific evidence may be strong, but the information is not standardized for database integration.

Each paper may use different:

- Glycan names
- Measurement methods
- Derived glycan traits
- Specimen descriptions
- Statistical results
- Disease terminology

Manual review would be needed to avoid treating broad glycan traits as exact glycan structures.

Data volume/scope estimate:

The publications page contains hundreds of listed research outputs spanning approximately 1991–2026.

However, the number of usable BiomarkerKB records cannot be estimated directly because the page does not identify which papers contain specific biomarker assertions. One publication may produce no usable records, while another may contain several separate glycan biomarkers.

Recommended integration status: Cross-reference

GlycanAge Publications is useful for finding high-value glycan biomarker studies, but it does not provide structured data that can be directly imported.

The most realistic use would be:

- Use the page as a curated bibliography.
- Select relevant open-access papers.
- Manually extract specific glycan biomarker findings.
- Map conditions, specimens, and glycan structures where possible.
- Add the publication as the evidence source.

A small manual curation project could be considered, but the publications page itself should not be treated as a database.

Open questions / limitations / blockers:

- Can GlycanAge or Genos provide a structured research dataset?
- Are exact glycan structures and external IDs available internally?
- How many publications contain usable biomarker-condition relationships?
- Which publications allow their text and supplementary data to be reused?
- Should aging itself be represented as a condition, phenotype, or separate biomarker role?

Field	Test/Example Filing 4: https://pubmed.ncbi.nlm.nih.gov/27302279/
Source record ID / persistent URL (biomarker_id)	DOI: 10.1038/srep28098
biomarker (CRITICAL)	decreased galactosylation of IgG glycans
assessed_biomarker_entity (CRITICAL)	Immunoglobulin G glycans
assessed_biomarker_entity_id (CRITICAL)	
assessed entity type	glycan
condition (CRITICAL)	colorectal cancer

condition id	
exposure agent	
exposure agent id	
best biomarker role	prognostic
specimen	plasma
specimen id	
loinc code	
evidence source	PMID: 27302279
evidence	Decreased galactosylation was associated with all-cause and colorectal cancer mortality.

Notes: GlycanAge is a publication collection rather than a structured biomarker database, so there is no GlycanAge biomarker record ID. I left the glycan ID blank because this publication page does not provide a GlyTouCan or other BiomarkerKB-compatible glycan identifier.

Evaluation report 5: Cancer Genome Interpreter Biomarkers

Arthur Issler

Source URL(s):

<https://www.cancergenomeinterpreter.org/>

<https://www.cancergenomeinterpreter.org/2021/biomarkers>

<https://www.cancergenomeinterpreter.org/2021/faq>

Maintainer/Organization: Cancer Genome Interpreter, or CGI, is developed by the Biomedical Genomics Lab at IRB Barcelona. The Cancer Biomarkers database was manually curated by clinical and scientific experts in precision oncology.

Last updated (if available): I did not find one confirmed overall update date. The downloaded table includes individual curation dates through April 21, 2024, although many records are older or have no curation date.

Description / what it covers:

CGI interprets genomic alterations found in tumors and identifies alterations that may affect response to cancer treatments.

Its Cancer Biomarkers database connects:

- Genomic alterations
- Cancer types
- Drugs or drug families
- Treatment response, resistance, or toxicity
- Clinical evidence levels
- Supporting publications, guidelines, or trials

The current CGI platform now relies on the VICC MetaKB for drug-response biomarkers. The standalone Cancer Biomarkers database remains available through the older CGI website.

Access method (API, bulk download, scrape, manual):

The standalone biomarker data can be accessed through: A searchable table on the legacy website, Manual browsing, A downloadable tab-separated file. The downloaded file contains 20 structured columns. CGI also has a REST API for submitting and downloading analysis results,

but I did not confirm a dedicated API for retrieving the complete standalone Cancer Biomarkers database.

License:

The standalone Cancer Biomarkers database is licensed under CC0 1.0, meaning its content is dedicated to the public domain. This is different from results generated by the CGI analysis tool, which are licensed under CC BY-NC 4.0. For this project, the CC0 license of the Cancer Biomarkers database is the relevant license.

Field mapping: can it populate the critical fields and the rest of the schema?:

Field	Support	Notes
Source record ID/persistent URL (biomarker_id)	No	The downloaded table does not include a stable record identifier. BiomarkerKB would need to create one.
biomarker (CRITICAL)	Yes	The Biomarker and Alteration columns provide specific alterations such as EGFR T790M, ERBB2 amplification, or ALK fusion.
assessed_biomarker_entity (CRITICAL)	Yes	Gene symbols are provided in the Gene column.
assessed_biomarker_entity_id (CRITICAL)	Mappable	The table provides gene symbols but no HGNC, NCBI Gene, or other external gene IDs.
assessed_entity_type	Partial / Derived	Alteration types such as mutation, copy-number alteration, fusion, expression, and biallelic inactivation are provided. Mapping to BiomarkerKB types such as DNA or RNA would require rules, especially for expression records.
condition (CRITICAL)	Yes	Full cancer names are provided in the Primary Tumor type full name column.
condition_id	Mappable	CGI provides cancer abbreviations and names, but not DOID or MONDO identifiers in the downloaded table.

exposure_agent	Partial	Most records provide a drug or drug combination. Some records provide only a drug family rather than a specific drug.
exposure_agent_id	Mappable	Drug names are provided without standard identifiers such as ChEBI, DrugBank, or RxNorm.
best_biomarker_role	Derived	The records are predictive biomarkers because they describe treatment response, resistance, lack of response, or toxicity. The role is not stored directly as predictive.
specimen	No	There is no dedicated specimen field. Assay type sometimes describes a model such as a cell line, but this is not a patient specimen.
specimen_id	No	No specimen ontology identifiers are provided.
loinc_code	No	No LOINC codes are provided.
evidence_source	Yes	Every reviewed row contains a source such as a PMID, FDA guidance, NCCN guidance, clinical trial, or conference abstract.
evidence	Partial	Evidence level and association are consistently included. Some records also contain useful comments, but a direct supporting text excerpt is not consistently available.

Data quality / standardization:

The data are structured and manually curated. Important fields such as biomarker, gene, alteration, treatment association, evidence level, source, and cancer type were populated in all 1,170 downloaded rows. However, integration would still require cleaning because:

- Gene, drug, and condition identifiers are missing.
- Some rows contain multiple genes, drugs, alterations, or cancer types.
- Drug and evidence formats are not completely consistent.
- Some older records contain spelling or formatting issues.
- Comments, assay types, and curation dates are often blank.

The source is strong for therapeutic interpretation but does not directly match the BiomarkerKB schema.

Data volume/scope estimate:

According to chatGPT the raw downloaded table contains:

- 1,170 biomarker-treatment records
- 513 unique biomarker descriptions
- 209 unique gene or gene-combination values
- 216 unique drug or drug-combination values
- 138 unique cancer-type descriptions

The main alteration categories are mutations, copy-number alterations, fusions, expression changes, and biallelic inactivation.

The treatment associations include:

- 909 responsive records
- 197 resistant records
- 38 no-response records
- 26 toxicity records

These counts describe the downloaded file and may differ from the content currently available through VICC MetaKB.

Recommended integration status: Cross-reference

CGI provides strong, structured cancer biomarker data and supports the critical BiomarkerKB fields. However, CGI is already included through VICC MetaKB, which BiomarkerKB currently integrates. Importing the standalone CGI dataset directly would likely create duplicate records. The best approach is to keep CGI as a cross-reference and compare a small sample against the MetaKB integration. If the standalone file contains useful details that MetaKB does not preserve, such as curator notes, drug status, curation dates, or older evidence, those fields could be added as annotations rather than re-importing the full dataset.

Open questions / limitations / blockers:

- How much of the standalone CGI dataset is already present in BiomarkerKB through MetaKB?
- Does MetaKB preserve CGI fields such as drug status, curator, curation date, and comments?

- Is the standalone Cancer Biomarkers table still actively maintained?
- How should rows containing multiple genes, drugs, or cancer types be separated?
- Which ontology mappings should be used for genes, drugs, and cancer types?
- How should No Responsive, Resistant, and different toxicity categories be standardized?
- Are any records dated after April 2024 available through CGI or MetaKB?

Field	Test/Example Filing 5: Cancer Genome Interpreter https://doi.org/10.34810/DATA405
Source record ID / persistent URL (biomarker_id)	
biomarker (CRITICAL)	presence of BRAF V600E mutation
assessed_biomarker_entity (CRITICAL)	BRAF
assessed_biomarker_entity_id (CRITICAL)	
assessed_entity_type	gene
condition (CRITICAL)	cutaneous melanoma
condition_id	
exposure_agent	vemurafenib
exposure_agent_id	
best_biomarker_role	predictive
specimen	
specimen_id	
loinc_code	
evidence_source	FDA
evidence	Responsive to vemurafenib; FDA guidelines

In the downloadable biomarkers.tab file, look for the row containing BRAF (V600E) + Vemurafenib + Cutaneous melanoma + FDA guidelines. The deposited dataset contains 999 biomarker records and is the archived Cancer Genome Interpreter Cancer Biomarkers database.

Notes: CGI does not give this row its own individual biomarker ID, so I left that blank. I used predictive because this specific entry connects the BRAF V600E mutation with response to a specific treatment, which directly fits the predictive role.

Evaluation report 6: CarboCurator

Arthur Issler

Source URL(s): <https://github.com/glygener/CarboCurator>

<https://github.com/issues/assigned?issue=clinical-biomarkers%7Cbiomarker-issue-repo%7C248>

Maintainer/Organization: CarboCurator is hosted under the GlyGen GitHub organization. The public repository identifies Cyrus C.H. Au Yeung as the project author.

Last updated (if available): Jun 16, 2025

Description / what it covers:

CarboCurator is not a public biomarker database. It is a text-mining pipeline that uses large language models to extract glycan biomarker relationships from PubMed abstracts.

The processed output connects:

- Glycan entities and changes
- Diseases
- Biomarker roles
- Specimens
- Medical interventions
- Glycoproteins and glycoenzymes
- PMIDs and supporting abstract sentences

The repository produces structured TSV and JSONL outputs that could potentially be converted into BiomarkerKB records.

Access method (API, bulk download, scrape, manual):

Access is available through:

- Public Python code on GitHub
- Processed TSV files in the repository
- Intermediate JSONL outputs
- Local execution of the pipeline

The existing TSV can be downloaded directly without rerunning the pipeline. Running the pipeline requires an OpenAI API key. I did not find a public API or searchable biomarker interface.

License:

The licensing status is unclear. The repository's license section currently states "Placeholder," and no completed license file is provided.

The code and processed data should therefore not be treated as openly reusable outside the project until the intended license is confirmed.

Field mapping: can it populate the critical fields and the rest of the schema?:

BiomarkerKB field	Support	Notes
Source record ID/persistent URL (biomarker_id)	No	No stable record ID is provided.
biomarker (CRITICAL)	Derived	A biomarker statement can be constructed from the glycan change, entity, disease, specimen, and role.
assessed_biomarker_entity (CRITICAL)	Partial	Glycan entities are usually present, but some are broad terms such as "glycans" or "glycosylation."
assessed_biomarker_entity_id (CRITICAL)	Partial/ Mappable	No GlyTouCan or other glycan identifiers are included.
assessed_entity_type	Derived	Most records could be classified as glycan, but some involve glycoproteins, enzymes, or general glycosylation traits.
condition (CRITICAL)	Partial	Disease names are present in approximately 83% of rows.
condition_id	Mappable	Mapped disease labels are provided, but DOID or MONDO identifiers are not.
exposure_agent	Partial	Medical interventions are present in approximately 21% of rows.
exposure_agent_id	No	No standard drug or intervention identifiers are included.
best_biomarker_role	Partial	Roles are usually present but include diagnostic, prognostic, predictive, monitoring, response, risk, and safety.

specimen	Partial	Specimen information is usually present but is not consistently standardized.
specimen_id	Mappable	Mapped labels are provided, but ontology identifiers are not.
loinc_code	No	No LOINC codes are included.
evidence_source	Yes	Every row contains a PMID.
evidence	Yes	Every row contains a supporting sentence from the abstract.

Data Quality / Standardization:

ChatGPT Analysis of the downloaded TSV found:

- The processed TSV contains:
- 6,728 extracted rows
- 17 columns
- 4,593 unique PMIDs

Field completeness is generally strong:

- Glycan entity: approximately 97%
- Specimen: approximately 97%
- Biomarker role: approximately 91%
- Disease: approximately 83%
- Medical intervention: approximately 21%

However, high completeness does not guarantee accurate extraction or mapping. Main issues include:

- Broad or nonspecific glycan terms
- No standard glycan, disease, drug, or specimen IDs
- Mixed or unsupported biomarker roles
- Proteins or glycoproteins appearing in glycan fields
- Incorrect mappings, including Humans appearing as a disease or specimen
- Evidence based only on abstracts
- No clearly reported validation accuracy for the final TSV

The dataset is therefore promising but still requires review and normalization.

Data volume/scope estimate:

The dataset covers 4,593 PubMed articles and includes biomarkers from cancer, infectious disease, metabolic disease, inflammatory disease, neurological disease, and other areas.

It includes human and animal studies, multiple specimen types, and diagnostic, prognostic, predictive, monitoring, response, risk, and safety relationships.

The final number of usable BiomarkerKB records will be lower after vague, incorrectly mapped, unsupported, or duplicate records are removed.

Recommended integration status: Sample integration

CarboCurator is a strong candidate for sample integration because it already produces structured glycan biomarker records that align closely with the BiomarkerKB schema.

However, direct integration is not recommended yet because the records were generated through an LLM pipeline, standard ontology identifiers are missing, several mapping errors are visible, the licensing status is unresolved, and complete manual validation has not been confirmed.

I recommend reviewing a sample of 50 to 100 records that contain:

- A specific glycan entity
- A clear disease
- A human specimen
- A diagnostic, prognostic, or predictive role
- A sentence that directly supports the relationship

The pilot should measure extraction accuracy, ontology mapping success, duplication with existing BiomarkerKB records, and the amount of manual correction required.

Open questions / limitations / blockers:

- Was the processed TSV manually reviewed?
- What validation results were obtained?
- What license applies to the code and generated data?

- Have any records already been integrated into BiomarkerKB or GlyGen?
- Can specific glycans be mapped to GlyTouCan?
- How should broad glycan traits and mixed biomarker roles be represented?
- Can incorrect disease and specimen mappings be corrected automatically?
- Should evidence be checked against full publications rather than only abstracts?
- Who is responsible for maintaining and rerunning the pipeline?

Field	Test/Example Filing 6: CarboCurator
Source record ID / persistent URL (biomarker_id)	Blank
biomarker (CRITICAL)	increased level of glycan sialyl Lewis X
assessed_biomarker_entity (CRITICAL)	Sialyl Lewis X
assessed_biomarker_entity_id (CRITICAL)	
assessed entity type	glycan
condition (CRITICAL)	pancreatic cancer
condition_id	
exposure_agent	
exposure_agent_id	
best biomarker role	diagnostic
specimen	urine
specimen_id	
loinc code	
evidence source	PMID:33135073
evidence	“Representative structures of glycans with elevated levels in cancer patients included the following: sialyl Lewis X.”

Exact entry used:

https://github.com/glygener/CarboCurator/blob/main/data/processed/biomarker_table/biomarker_table_20250331_4o-mini.tsv

Search the TSV for 33135073 or sialyl Lewis X to find the exact row. The processed CarboCurator dataset is the appropriate record-level source for this test.

Notes: The biomarker wording is simply constructed from the row's glycan_entity_change = increased level + glycan_entity = sialyl Lewis X. I left the IDs blank because this row does not provide a GlyTouCan ID, condition ontology ID, UBERON ID, or row-level record ID.

Evaluation report 7: Alliance of Genome Resources

Arthur Issler

Source URL(s):

- <https://www.alliancegenome.org/>
- <https://www.alliancegenome.org/search?q=N-glycan>
- <https://www.alliancegenome.org/swagger-ui/>
- https://www.alliancegenome.org/agr_simplemine.cgi
- <https://alliancemine.alliancegenome.org/alliancemine>

Maintainer/Organization: The Alliance of Genome Resources is a consortium that combines data from major model organism databases, including FlyBase, MGD, RGD, SGD, WormBase, Xenbase, and ZFIN, together with Gene Ontology information. It supports comparative genomics across model organisms and humans.

Last updated (if available): Alliance is actively maintained. The most recent confirmed release is version 9.0.0, released on May 14, 2026, with refreshed data from its model organism sources and additional technical updates.

Description / what it covers:

Alliance integrates information about:

- Genes, alleles, and variants
- Gene function and pathways
- Disease associations
- Phenotypes and expression
- Orthology across species
- Experimental models and supporting literature

It is not designed as a biomarker database. Searches for terms such as “N-glycan” mainly return genes involved in glycan synthesis or processing. For example, the results describe genes such as Mgat2 and ALG9, their functions, orthologs, and disease associations, rather than measurable glycan changes associated with diagnosis, prognosis, or treatment response.

Access method (API, bulk download, scrape, manual):

Alliance provides several structured access methods:

- Website search and record pages
- Public REST API
- Bulk downloads
- SimpleMine tab-separated exports
- AllianceMine queries and web services

SimpleMine can export gene identifiers, disease associations, expression, variants, interactions, and orthologs. AllianceMine also supports custom queries and client libraries. These methods are preferable to webpage scraping.

License:

AllianceMine states that Alliance operates under the Creative Commons Attribution 4.0 International license. Because Alliance combines information from several member databases and Gene Ontology, attribution and any source-specific reuse requirements should still be checked when redistributing bulk data.

Field mapping: can it populate the critical fields and the rest of the schema?:

BiomarkerKB field	Support	Notes
Source record ID/persistent URL (biomarker_id)	No	Alliance does not provide biomarker records or biomarker IDs.
biomarker (CRITICAL)	No	Gene–disease and gene-function relationships are provided, but not structured biomarker assertions.
assessed_biomarker_entity (CRITICAL)	Partial	Genes, alleles, and variants could serve as assessed entities, but Alliance does not identify them specifically as biomarkers.
assessed_biomarker_entity_id (CRITICAL)	Yes	Strong identifiers are available, including HGNC, MGI, FlyBase, SGD, NCBI Gene, Ensembl, and UniProtKB identifiers.

assessed_entity_type	Partial	Gene, allele, and variant types are identified, but they would need to be mapped to BiomarkerKB's controlled vocabulary.
condition (CRITICAL)	Yes	Disease associations and disease pages are available.
condition_id	Yes	Disease Ontology identifiers are used directly.
exposure_agent	No	Drugs and other exposures are not consistently represented as part of disease or biomarker assertions.
exposure_agent_id	No	No standard exposure-agent field is provided.
best_biomarker_role	Partial	Some disease annotations may contain marker-like qualifiers, but diagnostic, prognostic, and predictive roles are not consistently represented.
specimen	No	No routine biological specimen field is provided for disease associations.
specimen_id	No	No specimen ontology field is available for biomarker records.
loinc_code	No	LOINC codes are not provided.
evidence_source	Partial	Publications, source databases, and evidence codes are available for many annotations.
evidence	Partial	Annotation details and descriptions are provided, but not concise evidence statements supporting biomarker relationships.

Data quality / standardization:

Alliance is a high-quality resource for genes, variants, orthology, disease terms, and functional annotations. It uses strong identifiers and integrates information from established model organism databases. However, this information does not directly meet the BiomarkerKB definition of a biomarker record. A gene being involved in N-glycan processing or associated with a disease does not necessarily mean that the gene or glycan is a measurable diagnostic, prognostic, or predictive biomarker.

For example, the Mgat2 record describes its role in N-glycan diversification and its connection to congenital glycosylation disorders. This is useful biological context, but it does not state that a measurable change in Mgat2 or a specific glycan is a clinical biomarker

Data volume/scope estimate:

Alliance is a very large cross-species genomics resource covering genes, alleles, variants, diseases, models, expression, function, and orthology. An exact biomarker count is not applicable because Alliance does not maintain a biomarker record type. A search result count for “N-glycan” would also not represent usable biomarkers, since the results include genes and functional annotations rather than validated biomarker relationships.

Recommended integration status: Cross-reference

I recommend using Alliance as a cross-reference and enrichment source rather than importing it as a new source of BiomarkerKB records.

Alliance could strengthen existing BiomarkerKB entries by providing:

- Standard gene and variant identifiers
- Disease Ontology mappings
- Cross-species orthology
- Gene-function context
- Model-organism evidence
- Links to supporting publications and source databases

However, it should not be used to generate biomarker records automatically. It lacks the measurable biomarker statement, specimen, exposure, clinical role, and direct evidence relationship needed by BiomarkerKB. Importing general gene–disease associations as biomarkers would risk overstating the source evidence.

Open questions / limitations / blockers:

- Are Alliance gene and disease identifiers already available through existing BiomarkerKB sources?
- Would orthology or model-organism evidence add useful context to current BiomarkerKB records?

- Are any Alliance disease qualifiers specific enough to support a limited biomarker use case?
- What attribution is required when redistributing data from individual Alliance member databases?
- Would adding Alliance links provide enough value to justify maintaining another cross-reference?

Field	Test/Example Filing 7: Alliance of Genome Resources https://www.alliancegenome.org/gene/HGNC%3A31537
Source record ID / persistent URL (biomarker id)	HGNC:31537
biomarker (CRITICAL)	
assessed_biomarker_entity (CRITICAL)	MIR150
assessed_biomarker_entity_id (CRITICAL)	NCBI_Gene:406942
assessed_entity_type	gene
condition (CRITICAL)	colorectal cancer
condition id	
exposure agent	
exposure agent id	
best biomarker role	
specimen	
specimen id	
loinc code	
evidence_source	
evidence	Listed as a biomarker of colorectal cancer

Notes: The Alliance page directly identifies **MIR150**, gives **NCBI_Gene:406942**, and states that it is a biomarker of colorectal cancer. However, it does **not say what change or measurement of MIR150 is the biomarker**, so I left the critical biomarker field blank rather than turning MIR150 alone into a BiomarkerKB biomarker. It also does not provide a biomarker role, specimen, or supporting evidence on this entry page.